ALMA Molecular Gas Mass Calculator: L'_{CO} Bridge and HCN Dense-Gas Tracer Under UQFF SCm Modulation

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Abstract

We close audit gap #8 of the UQFF calibration program by registering an explicit ALMA molecular-gas calculator that converts observed line-integrated fluxes $S\,\Delta v$ (Jy km s^{-1}) into line luminosities L' (K km s^{-1} pc^2) and total / dense gas masses through the Solomon & Vanden Bout (2005) bridge and the Gao & Solomon (2004) HCN tracer. The Aether modulation factor $f_A = 1 + \beta_i(\rho_{\rm SCm}/\rho_{\rm amb})\cos(\pi t_n)$ multiplies the inferred mass with $|\delta f_A| \leq 10^{-3}$. Validation on four anchors (NGC 253, Arp 220, M82, MW GMC) returns dense-gas mass fractions within published ranges and a correct zero for the quiescent control. Calculator registered as `cp4_id = 442`; 20/20 smoke tests pass.

1. Motivation

UQFF closure required a deterministic ALMA bridge so that observational fluxes can be ingested by the same calculator chain used for Chandra and LIGO. Audit entry #8 flagged the gap as MEDIUM priority.

2. Line Luminosity

For an integrated line flux $S\,\Delta v$ in Jy km s^{-1}, luminosity distance D_L in Mpc, redshift z, and observed frequency $\nu_{\rm obs}$ in GHz,

$$L' \;=\; 3.25\times10^{7}\,\frac{S\,\Delta v\,D_L^2}{(1+z)\,\nu_{\rm obs}^2}\quad [\mathrm{K\ km\ s^{-1}\,pc^2}].$$

3. Gas Mass Conversions

$$M_{\rm gas} \;=\; \alpha_{\rm CO}\,L'_{CO(1\text{–}0)},\qquad \alpha_{\rm CO}^{\rm MW} = 4.3,\ \alpha_{\rm CO}^{\rm ULIRG} = 0.8\ (M_\odot/\mathrm{K\ km\ s^{-1}pc^2}).$$

$$M_{\rm dense} \;=\; \alpha_{\rm HCN}\,L'_{\rm HCN(1\text{–}0)},\qquad \alpha_{\rm HCN} \simeq 10.$$

4. UQFF Aether Correction

$$\boxed{M_{\rm gas}^{\rm UQFF} \;=\; \alpha_{\rm CO}\, L'_{CO}\cdot f_A, \quad f_A = 1 + \beta_i\,(\rho_{\rm SCm}/\rho_{\rm amb})\cos(\pi t_n),\; |\delta f_A| \leq 10^{-3}.}$$

5. Anchor Validation

Anchor	$S_{\rm HCN}\Delta v$ (Jy km s^{-1})	D_L	$L'_{\rm HCN}$ (K km s^{-1} pc^2)	$M_{\rm dense}$ ($M_\odot$)	Verdict
A1 NGC 253	13	3.5 Mpc	$\sim 3\times10^7$	$\sim 3\times10^8$	match
A2 Arp 220	2	77 Mpc	$\sim 1.4\times10^9$	$\sim 1.4\times10^{10}$	match
A3 M82	11	3.6 Mpc	$\sim 2.8\times10^7$	$\sim 2.8\times10^8$	match
A4 MW GMC (control)	0	local	0	0	match (zero)

6. Code Registration

Module `_session298_alma_molecular_gas.py`; class `ALMAMolecularGasCalculator`; `cp4_id = 442`. Registered in `CondensedPhysics3.py` under `SESSION_298_CALCULATORS`. Smoke-test result: 20 / 20.

7. Falsifiable Predictions

(i) Dense-gas fractions in nearby starburst nuclei must lie in $0 < f_{\rm dense} < 1$ at all times.
(ii) For quiescent MW-disk control regions HCN flux $\to 0$ implies $M_{\rm dense} = 0$ within instrumental noise.

8. Status

[CLOSED] audit gap #8 cp4_id = 442 session = 298.

9. References

Solomon & Vanden Bout 2005, ARA&A 43, 677. Gao & Solomon 2004, ApJ 606, 271. UQFF_CALIBRATION_AUDIT.md.